

LABORATORY OBSERVATION

Course: **Full Stack Web Development**

Course Code: **23CM4121**

Experiment No.: **1**

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Branch: **CSM**

Date: **30-08-2026**

1. TITLE

College Information Portal

2. AIM

To design and implement a structured HTML5 webpage titled **College Information Portal** using basic HTML tags, tables, images, forms, semantic elements, HTML5 input controls, audio, video, hyperlinks and lists.

3. OBJECTIVE

- Create a valid HTML5 document with headings, paragraphs and text formatting.
- Use ordered and unordered lists and hyperlinks.
- Create a student details table and insert images with alternate text.
- Design a student registration form using different input controls.
- Use HTML5 semantic elements such as header, nav, section, article and footer.
- Demonstrate audio, video and HTML5 input types such as email, date, number, color and range.

4. THEORY

HTML (HyperText Markup Language) is the standard markup language used to structure content on web pages. HTML5 provides the basic document structure and introduces semantic elements and new form controls. Semantic elements describe the meaning and role of content. Examples include header, nav, section, article and footer. Lists organize items, tables represent tabular data, forms collect user input, and multimedia elements embed audio and video. HTML5 input types such as email, date, number, color and range provide suitable controls and browser-supported validation or specialized interfaces.

5. ALGORITHM / PROCEDURE

1. Create a new folder for Task 1 and create an HTML file.
2. Add the HTML5 DOCTYPE, html, head and body structure.
3. Add college information using headings, paragraphs and text formatting tags.
4. Create ordered and unordered lists and add hyperlinks.
5. Create a student details table with Roll No, Name, Branch and Email.
6. Insert the college logo and a relevant campus image with alternate text.
7. Create the student registration form with the required controls.
8. Add semantic elements such as header, nav, section, article and footer.
9. Add audio and video elements and HTML5 input types.
10. Open the HTML file in a browser and verify the page and controls.

6. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>College Information Portal</title>
</head>
<body>
  <!--Headings and Paragraphs-->
  <h1>College Information Portal</h1>
  <h2>About Our Institute</h2>
  <p>Welcome to the official Information Portal</p>
  <p>Our college provides world-class education and fosters innovation among students</p>
  <!--Text Formatting-->
  <h2>Academic Excellence</h2>
  <p>We are proud to be ranked<b> Number 1</b> in the region. Our<i> State-of-the-art</i> campus offers a <u>dynamic learning</u>
  <p>Check out our award-winning formula :  $E=mc^2$ . Alternatively, look at our chemistry lab schedules for  $H_2$ 
  <!--Unordered List-->
  <h2>Available Courses</h2>
  <ul>
    <li>Computer Science Engineerinh</li>
    <li>Data Science</li>
    <li>Artificial Intelligence</li>
  </ul>
  <!--Ordered List-->
  <ol>
    <li>Fill out the online application form.</li>
    <li>Submit your Academic transcript.</li>
    <li>Attend the counseling session.</li>
  </ol>
  <!--HyperLinks-->
  <h2>Useful Links</h2>
  <p>Visit the official <a href="https://anits.org/" target="_blank">CollegeWebsite</a>for more details.</p>
  <p>For external research, you can search on <a href="https://www.google.com/" target="_blank">Google</a>.</p>
  <h2>Student Details</h2>
  <!--Student details table-->
  <table border="1">
    <tr>
      <th>Roll No</th>
      <th>Name</th>
      <th>Branch</th>
      <th>Email</th>
    </tr>
    <tr>
      <td>100</td>
      <td>Raj</td>
      <td>CSE</td>
      <td>raj@gmail.com</td>
    </tr>
    <tr>
      <td>101</td>
      <td>Priya</td>
      <td>CSM</td>
      <td>priya@gmail.com</td>
    </tr>
  </table>
  <!--Images-->
  <h2>College Logo</h2>
  
  <h2>Campus Image</h2>
  
  <!--Forms-->
  <h2>Student Registration Form</h2>
  <form>
    Name: <input type="text"><br><br>
    Email: <input type="email"><br><br>
    Password: <input type="password"><br><br>
    Gender: <input type="radio" name="Gender">Male
    <input type="radio" name="Gender">Female
    <br><br>
    Hobbies: <input type="checkbox">Sports
    <input type="checkbox">Reading
```

6. PROGRAM (continued)

index.html

```
<input type="checkbox">Music
<br><br>
Branch:
<select>
  <option>CSE</option>
  <option>ECE</option>
  <option>CSM</option>
  <option>CSD</option>
</select> <br><br>
Comments <br>
<textarea rows="4" cols="30"></textarea> <br><br>
<input type="submit">
<input type="reset">
</form>
<!--HTML5 Features-->
<header>
  <h1>College Information Portal</h1>
</header>
<nav>
  <a href="#">Home</a>
  <a href="#">About</a>
  <a href="#">Contact</a>
</nav>
<section>
  <article>
    <h2>About College</h2>
    <p>ANITS college provides quality education.</p>
  </article>
</section>
<h3>Audio</h3>
<audio controls>
  <source src="audio.mp3" type="audio/mpeg">
</audio>
<h3>Video</h3>
<video width="400"controls>
  <source src="video.mp4" type="video/mp4">
</video>
<form>
  Email: <input type="email"><br><br>
  Date: <input type="date"><br><br>
  Number: <input type="number"><br><br>
  Favorite Color: <input type="color"><br><br>
  Range: <input type="range"><br><br>
  <footer>
    <p>Copyright 2026 ANITS College</p>
  </footer>
</form>
</body>
</html>
```

7. CODE EXPLANATION

Code / Element	Explanation
<!DOCTYPE html>	Declares the document as HTML5.
<html>, <head>, <body>	Define the root element, metadata area and visible page content.
<h1>, <h2>, <h3>, <p>	Create headings and paragraphs for structured content.
, <i>, <u>, <sup>, <sub>	Apply bold, italic, underline, superscript and subscript formatting.
/	Create ordered and unordered lists.
<a>	Creates hyperlinks to the college website and Google.
<table>	Displays student information in rows and columns.
	Displays an image and provides alternative text.
<form>, <input>, <select>, <textarea>	Collect and control user input.
type="email"	Provides email-specific input and browser validation.
type="date"	Provides a date picker in supporting browsers.
type="number"	Provides numeric input.
type="color"	Provides a color picker.
type="range"	Provides a slider control.
<header>, <nav>, <section>, <article>	Separate HTML5 elements that describe the role of page content.
<audio> / <video>	Embed multimedia files in the webpage.

8. TEST CASES AND EXPECTED OUTPUT

Test Case	Input / Action	Expected Result	Status
TC-01	Open index.html	College Information Portal loads in the browser.	Pass
TC-02	Click College Website / Google	Selected link opens the target webpage.	Pass
TC-03	Enter email	Email input accepts a valid email format.	Pass
TC-04	Select gender	Radio buttons allow one gender option to be selected.	Pass
TC-05	Select hobbies	Multiple checkboxes can be selected.	Pass
TC-06	Use branch dropdown	Branch option can be selected.	Pass
TC-07	Use Date / Number / Color / Range	HTML5 controls are displayed and usable.	Pass
TC-08	Click Reset	Form values return to their default state.	Pass
TC-09	Play audio / video	Browser media controls are displayed.	Pass
TC-10	Incorrect image path	Image is not displayed; alt text may be shown.	Pass

9. OUTPUT

The webpage is expected to display the title **College Information Portal**, college information, formatted text, lists, hyperlinks, student table, images, registration form, semantic HTML5 sections, audio/video controls and modern HTML5 input controls. The actual multimedia files should be kept in the same folder as the HTML file using the filenames referenced in the code.

Required local files referenced by the program: logo.jpg, campus.jpg, audio.mp3 and video.mp4.

10. INTERVIEW / VIVA QUESTIONS

1. What is the purpose of semantic HTML5 elements?

Semantic elements communicate the meaning and role of content to browsers, developers and assistive technologies. They improve readability, accessibility and search-engine understanding. Examples are header, nav, section, article and footer.

2. Compare Ordered and Unordered Lists.

An ordered list () is used when the sequence matters and normally displays numbers. An unordered list () is used when the sequence does not matter and normally displays bullets.

3. What happens if an image path is incorrect?

The browser cannot locate the image resource, so the image is not displayed. If an alt attribute is provided, its alternative text can be shown and can also be used by assistive technologies.

4. Differentiate Radio Buttons and Checkboxes.

Radio buttons are normally used to select one option from a group and use the same name attribute. Checkboxes are independent controls and allow multiple selections.

5. Explain the advantages of HTML5 input types over traditional text boxes.

HTML5 input types provide specialized controls and built-in browser support. Email supports email-style validation, date provides a date picker, number supports numeric input, color provides a color picker, and range provides a slider. These controls improve usability and reduce the need for simple JavaScript validation.

Relevant code examples:

```
<header>College Information Portal</header>
<nav><a href="https://www.google.com/">Google</a></nav>
<input type="email">
<input type="date">
<input type="number">
<input type="color">
<input type="range">
<input type="radio" name="gender">
<input type="checkbox" name="hobbies">
```

11. RESULT

The College Information Portal was successfully designed using HTML5. The webpage demonstrates basic HTML structure, text formatting, lists, hyperlinks, tables, images, forms, semantic elements, HTML5 input types, audio and video embedding.

12. LEARNING OUTCOMES

- Develop well-structured HTML5 pages.
- Design tables and forms using appropriate controls.
- Use semantic elements and multimedia.
- Apply modern HTML5 input controls and browser validation.

9. OUTPUT

Actual browser output of the implemented HTML page:

College Information Portal

About Our Institute

Welcome to the official Information Portal

Our college provides world-class education and fosters innovation among students

Academic Excellence

We are proud to be ranked **Number 1** in the region. Our *State-of-the-art* campus offers a dynamic learning environment

Check out our award-winning formula : $E=mc^2$. Alternatively, look at our chemistry lab schedules for H_2O solutions.

Available Courses

- Computer Science Engineerinh
- Data Science
- Artificial Intelligence

1. Fill out the online application form.
2. Submit your Academic transcript.
3. Attend the counseling session.

Useful Links

Visit the official [CollegeWebsite](#) for more details.

For external research, you can search on [Google](#).

Student Details

Roll No	Name	Branch	Email
100	Raj	CSE	raj@gmail.com
101	Priya	CSM	priya@gmail.com

College Logo



Campus Image



Student Registration Form

Name:

Email:

Password:

Gender: ☐ Male ☐ Female

Hobbies: ☐ Sports ☐ Reading ☐ Music

Branch:

Comments

College Information Portal

[Home](#) [About](#) [Contact](#)

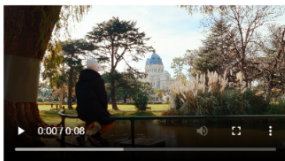
About College

ANITS college provides quality education.

Audio



Video



Email:

Date:

Number:

Favorite Color:

Range:

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